

Whether a Relation Extractor Learns Argument Order Depends on a Corpus Property Nobody Reports

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Abstract

A directional relation extractor should identify the same entity as subject whichever order the pair is presented in. Nothing in standard practice enforces this, and we show that whether it holds depends on a property of the training corpus that is never reported: the marginal probability that the first-listed argument is the subject. Across three corpora we find that when this prior is skewed (0.708) an unconstrained model takes the positional shortcut and its swap consistency falls to 0.473—below chance; when the prior is near balanced (0.486, 0.544) the model is forced to read content and reaches 0.94–0.97, but never exactness. The released state-of-the-art BioREDirect model sits at 0.700 on its own benchmark, where 17.9% of its directed predictions are returned unchanged under exchange—a self-contradiction—and 16.7% are abandoned to the undirected class. We give two mechanisms that make the property exact rather than contingent. Canonicalising argument order before encoding makes the two presentations byte-identical; an antisymmetric readout $s(h_A, h_B) = g(h_A, h_B) - g(h_B, h_A)$, paired with a *symmetric* directedness score, makes it hold at every parameter setting given order-invariant encodings. The first is what makes the property exact—to floating-point precision, whatever the head; the second recovers most of it (0.987) when the encoder stays order-sensitive. Together they cost 0.54% of parameters and 2% of throughput. The guarantee is free rather than beneficial: direction accuracy is statistically indistinguishable from the unconstrained arm on both public benchmarks (0.911 vs 0.912 on BioRED, $p = 0.79$; 0.963 vs 0.966 on SemEval-2010 Task 8, $p = 0.13$). We argue the right way to obtain a property the evaluation convention already assumes is to construct it, not to hope the corpus happens to supply it, and we release a swap-consistency protocol that needs no new annotation and runs on any existing extractor.

1 Introduction

For directional relations, the label encodes an ordering: *Cause-Effect*($e1, e2$) and *Cause-Effect*($e2, e1$) are distinct classes, and confusing them produces a knowledge-base edge pointing the wrong way. The convention is near-universal, from SemEval-2010 Task 8 (Hendrickx et al., 2010) through TACRED (Zhang et al., 2017) and DocRED (Yao et al., 2019) to recent biomedical work (Lai et al., 2025).

The convention carries an unstated assumption. The same pair can be presented in either order, and the two presentations are semantically identical: if A causes B then presenting (B, A) must yield the mirrored label, so that the entity identified as subject does not change. Nothing in the standard training objective enforces this. Each pair is seen in one order during training and the model is never asked what it would say about the reverse.

We find that the property is therefore not reliably learned—and, more interestingly, that whether it is learned at all depends on a property of the corpus that is never reported. Write π for the marginal probability that the first-listed argument is the subject. Across three corpora with $\pi = 0.708$, 0.544 and 0.486, an unconstrained model trained on order-sensitive input reaches swap consistency 0.473, 0.938 and 0.969 respectively (Figure 1). When π is skewed the model takes the positional shortcut and performs *below chance* on consistency; when π is balanced it is forced to read content and approximates the property without ever attaining it. The released BioREDirect model (Lai et al., 2025) sits at 0.700 on its own benchmark.

This reframes what the architecture is for. The property is not merely unenforced; it is contingent on a corpus statistic that practitioners neither measure nor control. Constructing it removes the contingency. We give two mechanisms that act at different points. The first removes argument order from the encoder: sorting the two en-

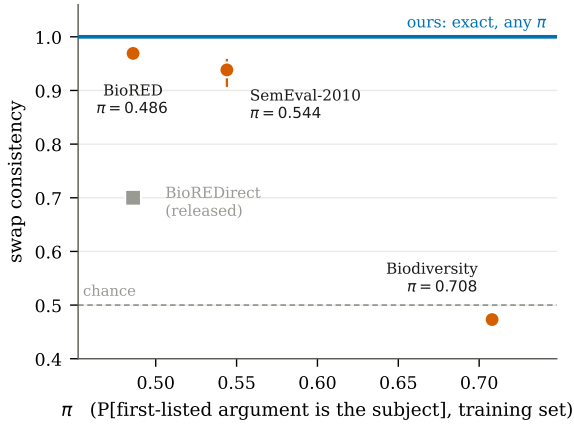


Figure 1: Swap consistency of an unconstrained model on order-sensitive input, against the training corpus’s positional prior π . The property is acquired only when the corpus withholds the positional shortcut; at $\pi = 0.708$ it falls below chance. Bars span three seeds (biodiversity is a single run). Our canonical arm is at 1.000 for every π , by construction rather than by training.

ties canonically before marking them makes the two presentations byte-identical, so consistency holds exactly, for any readout. The second constrains the readout: an antisymmetric head returns $s(h_A, h_B) = -s(h_B, h_A)$ at every parameter setting, which delivers the property whenever the encodings it is given are themselves order-invariant, and recovers most of it (0.987) when they are not. We pair the latter with a *symmetric* directedness score, because whether a relation is directed at all must *not* change under exchange—a distinction a single softmax over {rightward, leftward, undirected} cannot express.

Our claims, each supported on more than one corpus:

1. **Argument-order equivariance is contingent, not guaranteed**, and the contingency is predictable from a measurable corpus statistic (§4.1).
2. **Exactness is cheap**. 0.54% additional parameters, 2% of throughput, holding to floating-point precision rather than approximately (§4.3).
3. **It is free**. Direction accuracy is statistically indistinguishable from the unconstrained arm on both public benchmarks (§4.4). We claim no accuracy benefit.

We are explicit about what is not new. Separating relation identity from relation direction dates to Rink and Harabagiu (2010). Antisymmetry by construction is standard in preference learning

(Burges et al., 2005; Rigutini et al., 2011) and motivates several knowledge-graph embedding models (Trouillon et al., 2016; Sun et al., 2019). Placing a direction head on a shared encoder is Lai et al. (2025). What we have not found is an argument-order-invariant *input* for relation extraction: the field puts order into the label space, into role markers (Zhou and Chen, 2022; Baldini Soares et al., 2019), or into a constrained permutation set (Woo et al., 2022), rather than removing it from the encoder and recovering it at an equivariant readout.

2 Literature review

Direction in relation extraction. The dominant treatment folds direction into the label set. SemEval-2010 Task 8 (Hendrickx et al., 2010) defines nine relation types in two argument orders plus *Other*; TACRED (Zhang et al., 2017) and DocRED (Yao et al., 2019) order the arguments in the instance instead. The winning SemEval system (Rink and Harabagiu, 2010) already factored the problem, using one classifier for relation type and a second, per type, for direction, so the factorisation itself is not novel. What it lacks, and what we add, is any invariance property: both stages see the arguments in text order and are free to exploit it.

Entity markers. Marking the arguments in the passage is standard and the punctuation we use is borrowed. Zhou and Chen (2022) establish the typed entity marker in which the subject is enclosed in @ and the object in #; Baldini Soares et al. (2019) popularised the idea. We invert the semantics: our markers denote *positions in a canonical ordering*, not roles. A reader familiar with the convention will otherwise assume the standard reading, under which the invariance argument does not hold, so we flag the inversion explicitly.

Symmetry as an architectural constraint. Enforcing $f(a, b) = -f(b, a)$ by construction is old. RankNet (Burges et al., 2005) gives $P(a \succ b) = 1 - P(b \succ a)$ identically; SortNet (Rigutini et al., 2011) enforces the antisymmetry exactly by dual-neuron weight sharing and proves the constrained network remains a universal approximator—a stronger result than the difference form we use. In knowledge-graph embedding, the inability of symmetric scoring functions to represent antisymmetric relations motivates ComplEx (Trouillon et al., 2016) and RotatE (Sun et al., 2019). Our contribution is not the construction but its application:

to a text encoder that must read direction from a passage, paired with a symmetric score for undirectedness, and measured.

Invariance and equivariance. Woo et al. (2022), in scene-graph generation, state the goal in the same vocabulary—equivariant under subject-object ordering while invariant to the permutation—and resolve it in the opposite direction, by constraining the permutation set so the subject always precedes the object. Frame averaging (Puny et al., 2022) is the general umbrella: our canonical sort is a canonicalisation frame giving exact invariance, and our antisymmetric readout is group averaging in the sign representation of S_2 .

Shortcut learning and dataset artefacts. That models exploit superficial statistics rather than the intended signal is well documented (Gururangan et al., 2018; McCoy et al., 2019; Geirhos et al., 2020). Our π statistic is an instance: a positional regularity that is invisible in standard reporting and that determines whether an intended invariance is acquired. Partial-input baselines (Poliak et al., 2018) are the established diagnostic for the analogous problem in inference; our masked-passage control (§4.4) plays that role here.

Consistency probes. Perturbing an input in a way that should have a predictable effect is established methodology (Ribeiro et al., 2020; Elazar et al., 2021). Yuan and Vlachos (2025) show pre-trained encoders are at chance on a symmetry probe built by swapping subject and object in Wikidata triples, and address it with symmetry-aware training objectives; we address it architecturally, so no training signal is required. Woo et al. (2022) introduce a bidirectional diagnostic in vision. Swap consistency (§3.3) is the analogue for directional relation extraction and needs no new annotation.

Biomedical directional extraction. Lai et al. (2025) is our principal baseline. It adds 10,864 manual directionality annotations to BioRED (Luo et al., 2022) and predicts relation type, novelty and direction from three heads on a shared encoder, with soft prompts and context chunking. Its direction head is a four-way linear classifier over $\{None, Left_to_Right, Right_to_Left, No_Direct\}$ —confirmed here from the released weights—and carries no symmetry constraint.

3 Method

3.1 Task and the property at issue

Given a passage p , an ordered pair of marked mentions (x, y) and a relation type t , predict whether the relation is directed and, if so, which of x, y is its subject. Write $\sigma(p, x, y, t) \in \{x, y, \emptyset\}$. The property is

$$\sigma(p, x, y, t) = \sigma(p, y, x, t), \quad (1)$$

a statement about the identity of the predicted subject, not about the label: the label must mirror precisely so that the identified entity does not change.

3.2 The corpus statistic π

Let π be the fraction of directed training instances whose *first-listed* argument is the subject. A model given order-sensitive input can always achieve accuracy $\max(\pi, 1 - \pi)$ by ignoring the passage. Where π departs from 0.5 this shortcut is rewarded, and the intended content-based solution is correspondingly less necessary. π is trivially computable and, to our knowledge, never reported.

3.3 Swap consistency

We measure Eq. 1 directly: construct the exchanged presentation of each test pair, run the model on both, and record whether the same entity is named subject. The construction is mechanical but must exchange *every* slot in which the input encodes order—role markers and any template stating the pair order.

It is easy to get wrong. Our first attempt on BioREDirect exchanged only the source and target markers, leaving the question order intact, and reported 0.175. That is an artefact: the model reads the textual order of the entities rather than the marker labels, so the manipulation changed the gold answer without changing the argument order of the input. The corrected protocol exchanges both and gives 0.700. We report the error because it is the first thing a careful reader will probe.

3.4 Two routes to exactness

Let h_A, h_B be the contextual representations of the marked spans and c the sentence representation.

Canonical input (invariance). Sort the two entity strings by a fixed total order—case-folded lexicographic—and mark the first @ · @ and the second # · # in place. The token sequence is then a

function of the *unordered* pair: the two presentations are byte-identical and Eq. 1 holds for any readout. Decoding to a label happens outside the model, by comparing the @-marked entity to the first argument as given. Canonicalisation also neutralises π by construction, taking it to 0.5 regardless of the corpus.

Antisymmetric readout (equivariance). Independently of the input,

$$s_{\text{dir}} = g([h_A; h_B; c; e_t]) - g([h_B; h_A; c; e_t]), \quad (2)$$

$$s_{\text{und}} = h([h_A; h_B; c; e_t]) + h([h_B; h_A; c; e_t]), \quad (3)$$

with e_t a small embedding of the relation type and g, h two-layer networks. Eq. 2 is exactly antisymmetric and Eq. 3 exactly symmetric, at every parameter setting, before and during training.

The pairing carries the argument. *Which* entity is the subject must invert under exchange; *whether the relation is directed at all* must not. A single softmax over {rightward, leftward, undirected} expresses neither constraint: exchanging the arguments can move mass between rightward and leftward *and* into or out of undirected, so the model may hold two incompatible beliefs about one pair. We verify both properties numerically before training; residuals are 0 to machine precision.

Keeping the relation out of the encoder. The relation type enters only at the head, through e_t ; the encoder never sees the relation string. This makes the invariance argument airtight—direction cannot re-enter through the relation surface form—and lets one small head serve every relation in the schema.

3.5 Model, cost and protocol

The encoder is BiomedBERT-base (Gu et al., 2021) in all arms. The direction head is 594,465 parameters, 0.54% of the encoder, costing 2% of CPU throughput (33.0 against 33.8 pairs/s on eight threads). Arms within a comparison share encoder, optimiser, schedule, epochs and data, differing only on the axis under test. We report means over three seeds.

3.6 Corpora

BioRED (Luo et al., 2022) with the directionality annotations of Lai et al. (2025): 1,163 test pairs carrying a relation, 1,045 directed (527/518) and 118

Corpus	π	Uncon. swap	Ours
Biodiversity	0.708	0.473	1.000
SemEval-2010	0.544	0.938	1.000
BioRED	0.486	0.969	1.000
BioREDirect (released)		0.700	—

Table 1: Swap consistency of the order-sensitive unconstrained arm against the training-corpus prior π , with our canonical arm for reference.

undirected; $\pi = 0.486$. Clean mentions and passages are recovered from the PubTator source, using the smallest sentence window containing both entities as the baseline’s own chunking does.

SemEval-2010 Task 8 (Hendrickx et al., 2010): 7,991 training and 2,713 test instances after removing pairs with identical mention strings; 2,260 directed test instances and 453 *Other*, treated as undirected. General domain. $\pi = 0.544$.

Biodiversity verification set: 437 expert-graded candidate triples from a deployed literature-mining pipeline, plus 84 human-annotated direction items over 79 relation surface forms. $\pi = 0.708$ —the skewed case. Small, and treated accordingly.

4 Results

4.1 Equivariance is contingent on π

Table 1 and Figure 1 are the central result. An unconstrained model on order-sensitive input—standard practice—acquires the equivariance only when the corpus forces it to. At $\pi = 0.708$ the positional shortcut is available and consistency falls to 0.473, below chance. At $\pi \approx 0.5$ the shortcut is worthless and consistency reaches 0.94–0.97, but never 1.

We derived this relationship from the biodiversity and BioRED corpora and used it to predict SemEval before running it: at $\pi = 0.544$, mildly skewed, consistency should fall between and nearer the balanced end. It does.

4.2 What the failure looks like

On the released BioREDirect model, of the 854 BioRED test pairs for which it predicts a direction, 65.3% are correctly mirrored under exchange, 17.9% are returned unchanged—naming a different entity as subject depending on presentation order—and 16.7% are abandoned to the undirected class.

The failure is specific. Relation detection barely moves under exchange (F_1 0.7926 $\rightarrow$ 0.7861) while relation-plus-direction falls 0.5550 $\rightarrow$

0.4013. What the model knows about *whether* two entities are related survives; what it knows about *which way* does not.

4.3 The guarantee holds exactly

Across every arm using canonical input, swap consistency is 1.000 and the maximum deviation between a prediction and its mirror is 2.98×10^{-8} —floating-point precision, not approximation. This holds whichever head sits on top, because the encoder receives identical bytes in both presentations.

The antisymmetric readout does not by itself deliver the same thing, and it is worth being exact about why. Eq. 2 is antisymmetric in its two arguments at every parameter setting, so it guarantees the property *conditional on* h_A and h_B being unchanged by the exchange. An order-sensitive encoder breaks that condition: it genuinely sees different tokens when the markers move, so h_A and h_B are not the same vectors. Empirically the head recovers most of the gap but never closes it— 0.987 ± 0.002 on SemEval against 0.938 ± 0.028 for the unconstrained head, and 0.969 on the biodiversity set against 0.473. The two mechanisms are therefore not interchangeable: canonicalisation is what makes the property exact, and the antisymmetric head is what makes it nearly free to approximate when canonicalisation is unavailable—for instance where argument order carries information the task needs.

4.4 Accuracy: the guarantee is free

Table 2 gives the head-to-head. On both public benchmarks the canonical-symmetric arm and the order-sensitive unconstrained arm are indistinguishable. Summarising each arm by the majority vote of its three seeds and comparing per item, the unconstrained arm is nominally ahead on both—by 0.002 on BioRED (6 vs 8 discordant items, McNemar with continuity correction $p = 0.79$) and by 0.004 on SemEval (13 vs 23 discordant, $p = 0.13$)—and neither gap approaches significance. **We claim no accuracy benefit from the construction;** we claim it costs nothing.

The 0.766 of the released BioREDirect model is not directly comparable to either arm: it performs a joint task (relation, novelty, direction) with soft prompts and multi-chunk pooling, so the gap reflects system configuration rather than the design axis under test. We report it for context, not as evidence for our design.

Two controls support that the models read the

Corpus	Arm	Dir. acc.	Swap
BioRED	BioREDirect (released)	0.766	0.700
	text + unconstrained	0.912	0.969
	canonical + sym.	0.911	1.000
SemEval	text + unconstrained	0.966	0.938
	canonical + sym.	0.963	1.000

Table 2: Direction accuracy on gold-directed pairs and swap consistency. All BioRED and SemEval figures are means over three seeds; per-seed values and standard deviations are in the supplement.

passage rather than exploiting surface regularities. Withholding the relation type entirely changes BioRED accuracy from 0.909 to 0.910, so the head does not lean on that feature. Masking the passage and presenting only the two entity names drops the biodiversity direction model to 0.464—chance—so it is not recalling which entities typically stand in which role.

4.5 Decomposing the two mechanisms

Table 3 isolates each design choice, on the skewed biodiversity set where the distinction bites and on SemEval, where three seeds per cell and a 453-instance undirected class make the comparison stable.

The design separates into two main effects with no interaction. **Consistency is decided by the input format:** both canonical cells are exactly 1.000, and neither text cell reaches it whatever the head. **Undirectedness is decided by the head:** on SemEval the symmetric branch gains ≈ 0.043 AUC under both input formats. **Accuracy is decided by neither:** the four SemEval cells span 0.0036, within about two standard deviations of seed noise, and on biodiversity no difference reaches significance at $n = 84$ (canonical input $p = 0.239$; symmetric head $p = 1.000$).

One result does not replicate. On the 84-item set the unconstrained head looks like the better undirectedness detector (0.811 vs 0.748); on SemEval that reverses cleanly and consistently. The biodiversity set has too few undirected items to support the comparison, and we report the disagreement rather than the convenient half of it.

4.6 The applied setting

On the biodiversity verification set the same model raises F_1 from 0.816 to 0.902 against a deployed sentence-level filter’s own recorded decisions (McNemar $p = 2.3 \times 10^{-4}$, 437 items) at unchanged

Input	Head	Acc.	Und. AUC	Swap
<i>Biodiversity</i> , $n = 84$, $\pi = 0.708$, one run				
canon.	sym.	0.869	0.748	1.000
canon.	uncon.	0.857	0.787	1.000
text	sym.	0.810	0.718	0.969
text	uncon.	0.786	0.811	0.473
<i>SemEval</i> , $n = 2,260$, $\pi = 0.544$, 3 seeds				
canon.	sym.	0.9625	0.947	1.0000
canon.	uncon.	0.9634	0.904	1.0000
text	sym.	0.9661	0.947	0.9869
text	uncon.	0.9658	0.907	0.9383

Table 3: Design ablation on both corpora. Canonical input fixes swap consistency at floating-point exactness; the symmetric head fixes undirectedness detection; neither moves accuracy. SemEval figures are means over three seeds (standard deviations in the supplement).

inference cost. The honest characterisation is robustness to prevalence shift rather than raw accuracy: across blocks spanning prevalence 0.31 to 0.85, one fixed threshold gives an F_1 spread of 0.070 against the filter’s 0.169.

5 Discussion and conclusion

The measurement is the part we would most like reused. It costs nothing—no annotation, no retraining, one extra forward pass over an existing test set—and on the evidence here it is worth running on any released directional extractor.

The finding it produces is not that published systems are careless. It is that a property the evaluation convention silently assumes is delivered only when the corpus happens to make the alternative unprofitable. π is trivial to compute, is not reported in any dataset paper we examined, and predicts whether the intended invariance will be acquired. A corpus with a skewed positional prior will produce models that look accurate and are incoherent under exchange; a balanced one will not, and neither the model card nor the leaderboard will distinguish them.

The architectural response is modest in isolation and we have said so. The factorisation is [Rink and Harabagiu](#)’s; the antisymmetric construction is [Rigutini et al.](#)’s; the markers are [Zhou and Chen](#)’s with the semantics inverted. What is new is the combination, the measurement that motivates it, and the observation that the property is otherwise left to chance.

Two findings cut against a clean story and belong in the record. The unconstrained head detects undirectedness *better* than ours in three of four ab-

lation cells, peaking exactly where it is worst on consistency: the symmetry constraint costs flexibility in the undirected class. And on the biodiversity set no accuracy difference between design cells is significant—only the consistency differences are structural.

Limitations. The human direction gold for the applied setting is 84 items: enough to show the head reads the passage, not enough to separate it from a good dependency-parse rule, which we estimate would need on the order of 2,700 annotations. BioREDirect performs a joint task while our arms perform direction given the pair; we control for the most obvious confound (§4.4) but not all, and we therefore do not treat that comparison as evidence for the design. Our context window is the infix span where the baseline uses prefix, suffix and infix with max-pooling. The π relationship rests on three corpora; it is a mechanism with a plausible account and one successful prediction, not a law. Finally, the curated relation vocabulary shows that many relation surface forms carry no direction at all—12 of 79 annotated forms were marked undirected by the curator—so direction is defined only on a subset of extracted edges, and a system reporting it everywhere is overclaiming.

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A Per-seed values

Table 4 gives every seed behind the SemEval means in Table 3. Swap consistency for the two canonical cells is exactly 1.0000 in every run; the largest deviation between any prediction and its mirror is 2.98×10^{-8} across all six, which is the resolution of the float comparison rather than a measured discrepancy. The undirectedness AUC carries the largest seed variance of any quantity we report ($\sigma \approx 0.02-0.03$), which is why the head effect is stated as a difference of means over three seeds rather than from a single run: on seed 1 alone it would have read as 0.98 against 0.93, roughly twice its stable size.

Input	Head		s1	s2	s3
canon.	sym.	acc	0.9650	0.9615	0.9611
		swap	1.0000	1.0000	1.0000
		AUC	0.9823	0.9245	0.9354
canon.	uncon.	acc	0.9642	0.9624	0.9637
		swap	1.0000	1.0000	1.0000
		AUC	0.9297	0.8865	0.8951
text	sym.	acc	0.9668	0.9681	0.9633
		swap	0.9894	0.9863	0.9850
		AUC	0.9794	0.9238	0.9365
text	uncon.	acc	0.9642	0.9673	0.9659
		swap	0.9588	0.9066	0.9496
		AUC	0.9298	0.8952	0.8962

Table 4: SemEval-2010 Task 8, all three seeds per cell.

Arm		s1	s2	s3
canon. + sym.	acc	0.9091	0.9100	0.9129
	swap	1.0000	1.0000	1.0000
text + uncon.	acc	0.9120	0.9129	0.9100
	swap	0.9809	0.9694	0.9569

Table 5: BioRED test, 1,045 gold-directed pairs, all three seeds.